MOTION → PICK → ALIGN → RECONSTRUCT → ENHANCE → INTERPRET

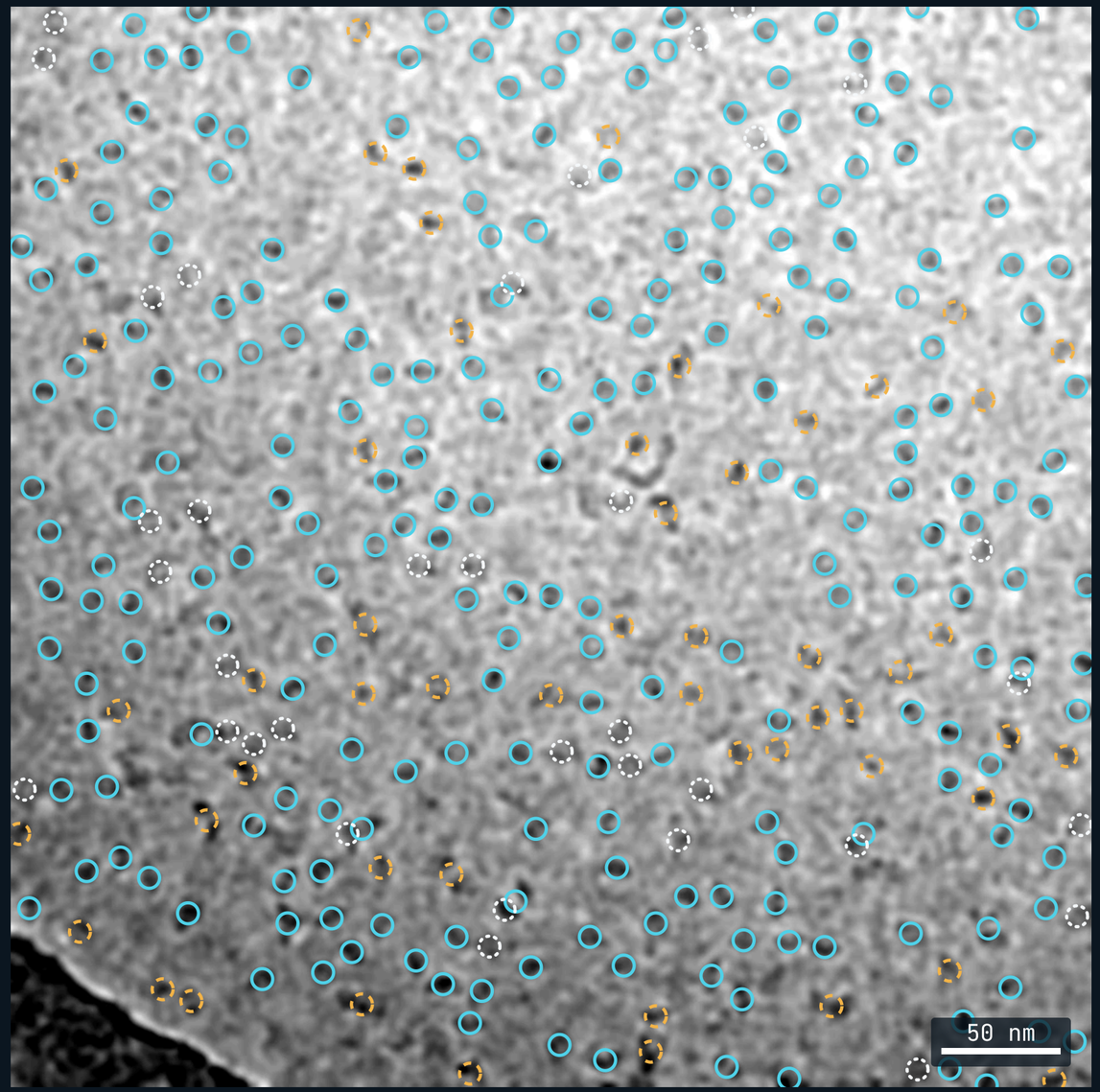
BCH 561 · BIOCHEMICAL TECHNIQUE SEMINAR

AI in cryo-EM and cryo-ET data processing

Ditsa Mallick · BCH 561 · Arizona State University · 27 October 2026

> Where does deep learning add **trustworthy** structural information — and how can we tell whether an apparent improvement is **real**?

● EXPERIMENTAL DATA ◆ RECOMPUTED RESULT



○ Accepted — model and expert agree (solid) ◌ Uncertain — model only (dashed)

◌ Missed — expert only (dotted)

SOURCE Micrograph: EMPIAR-10017 (β-galactosidase, Falcon II, 1.77 Å/px; expert coordinates by R. Henderson). Picks: Topaz 0.3.20 pretrained ResNet16, run by presenter.

" REPORTED RESULT ▢ CONCEPTUAL SCHEMATIC

Why AI-assisted processing matters

24 $e^{-}/Å^{2}$

total dose in the micrograph you just saw — every particle sits in noise

EMPIAR-10017

1,879

public EMPIAR datasets (4.83 PB) by 14 Jan 2025

Iqbal et al. 2026

437,235

particles picked by one tool on one dataset (Topaz, EMPIAR-10093)

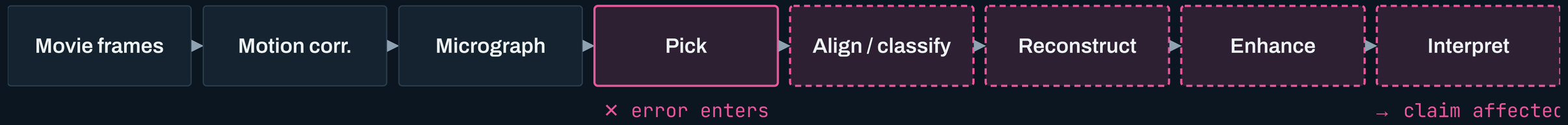
Dhakal et al. 2025, Table 3

≈570

particles per micrograph picked by hand by an expert (6,829 in 12 micrographs)

EMPIAR-10017 coordinates

SPEED IS NOT THE GOAL — ERRORS PROPAGATE



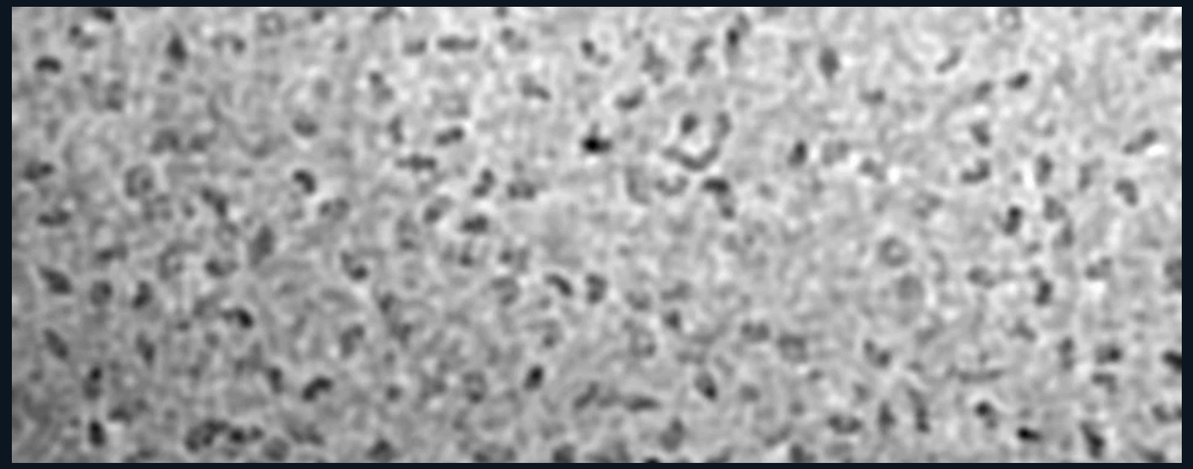
A wrong pick, a bad drift estimate or an over-sharpened map does not stay local: it reaches the structure and the biological claim.

SOURCE Iqbal et al. 2026 BMC Artif Intell 2:2 (EMPIAR size, Jan 2025) · Dhakal et al. 2025 Brief Bioinform 26:bbaf011 Table 3 · EMPIAR-10017 metadata

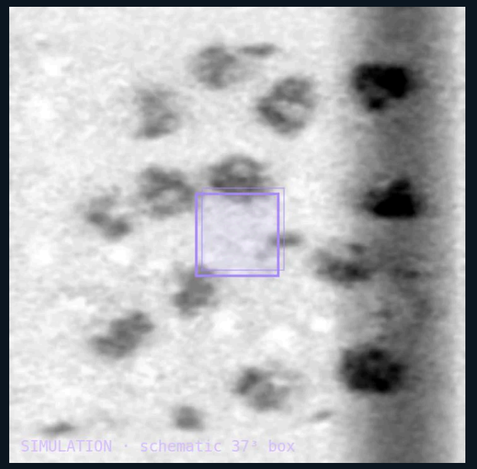
● EXPERIMENTAL DATA ◇ SIMULATION

Two branches: single particles vs. tomography

SINGLE-PARTICLE ANALYSIS · EXPERIMENTAL (EMPIAR-10017)



CRYO-ET · SIMULATION (PDB 6CVM + 4V6X, ±60° TILT, MEMBRANE SLAB)



	SINGLE-PARTICLE ANALYSIS (SPA)	CRYO-ELECTRON TOMOGRAPHY (CRYO-ET)
Sample	purified complex, thin ice	cells, lamellae, organelles
Measured	2D projections, one per particle	tilt series of one region (≈ ±60°)
Pick	2D particle coordinates	3D subtomogram coordinates
Average	single-particle reconstruction	subtomogram averaging
Typical result	often near-atomic	lower resolution, native context

What is the **atomic mechanism** of this purified complex?

Where is it, what is it bound to, what **state** is it in — inside the cell?

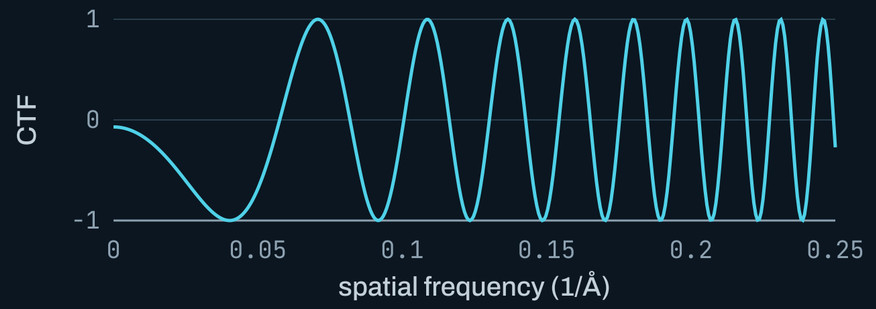
SOURCE Left: EMPIAR-10017 (experimental). Right: simulated tomogram slice from PDB 6CVM + 4V6X, ±60° tilt, presenter's deterministic simulation (seed 561). Iqbal et al. 2026.

◇ SIMULATION

Image formation: one equation, three unknowns

$$y_i = \mathrm{CTF}_i * \mathcal{P}_{\phi_i} V + n_i$$

y_i	particle image i (measured)
V	3D density of the molecule (unknown)
ϕ_i	its orientation and position (unknown)
$\mathcal{P}$	projection along the beam
CTF_i	microscope contrast transfer (estimated)
n_i	shot noise (unknown, large)



300 kV, Cs 2.7 mm, defocus 1.5 µm. Varying defocus moves the zeros.

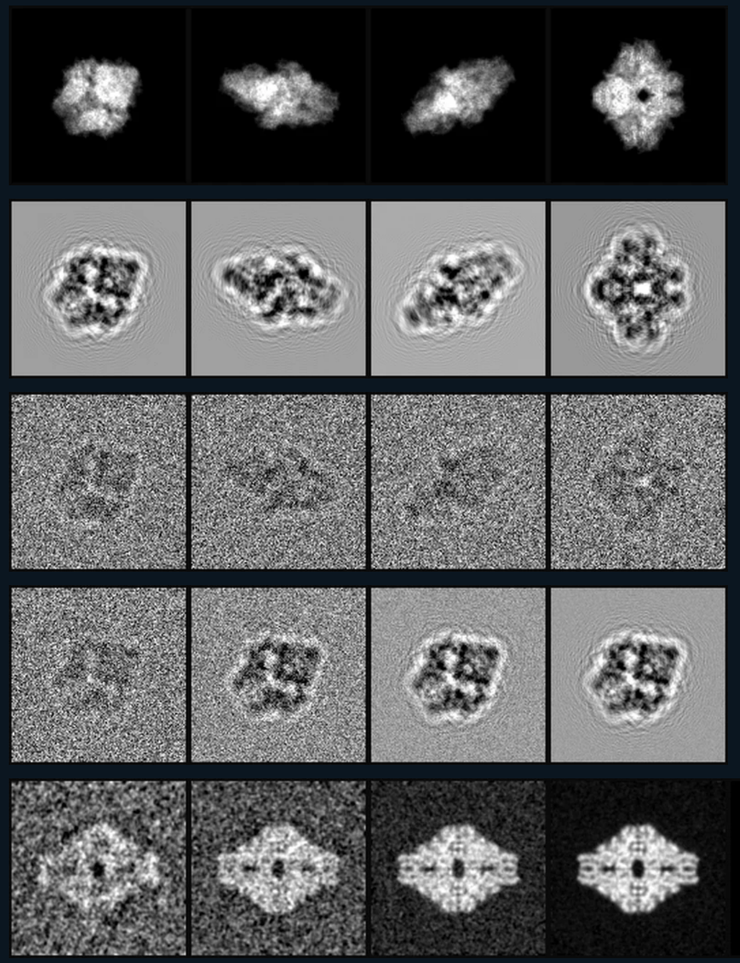
1 · Projection
random orientations φ, summed along the beam

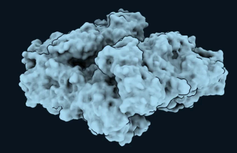
2 · × CTF
defocus-dependent contrast flips and zeros

3 · + noise n
SNR 0.05 — noise ≫ signal at a safe dose

4 · Average N aligned copies
N = 1, 10, 100, 1000

5 · Many views → 3D
central slice from 30 / 300 / 3,000 / 30,000 views, then truth





V: β-galactosidase
(PDB 6CVM, simulated)

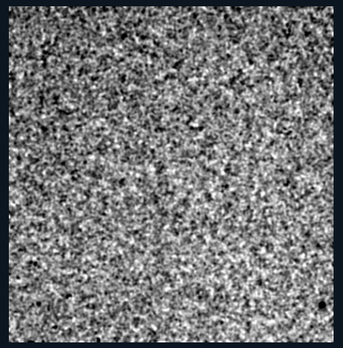
SOURCE Simulation from PDB 6CVM (β-galactosidase) via ChimeraX 1.12 molmap 4 Å; 300 kV, Cs 2.7 mm, 7% amplitude contrast, 1.5 µm defocus, SNR 0.05; known poses; seed 561.

● EXPERIMENTAL DATA ◆ RECOMPUTED RESULT ◇ SIMULATION

What the raw data look like

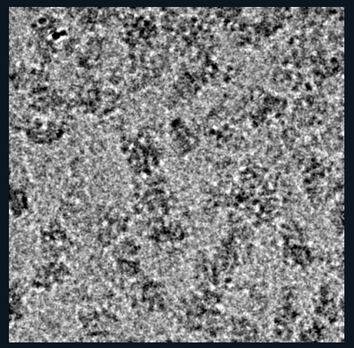
MEASURED

1 MOVIE FRAME (OF 49)



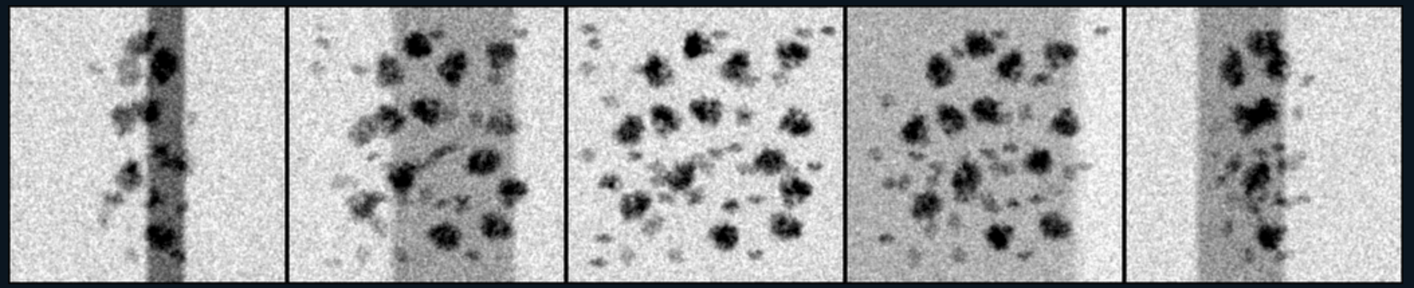
EXPERIMENTAL · EMPIAR-10204 · ≈1.3 $e^{-}/Å^{2}$

ALIGNED FRAME SUM = MICROGRAPH



RECOMPUTED from the 49 frames

TILT SERIES -60° … +60°

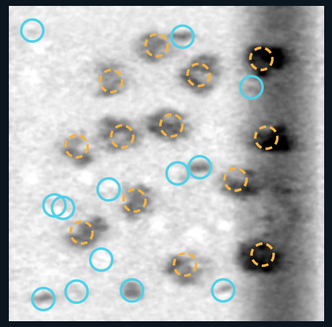


SIMULATION · PDB 6CVM + 4V6X

▲ measured · inferred ▼

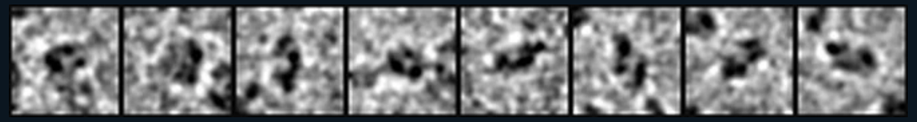
INFERRED

TOMOGRAM + 3D COORDINATES



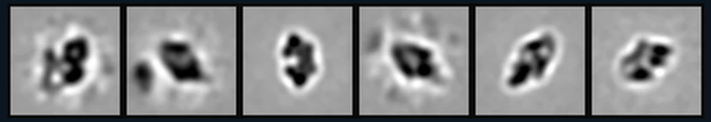
SIMULATION · ground-truth positions

EXTRACTED PARTICLE BOXES



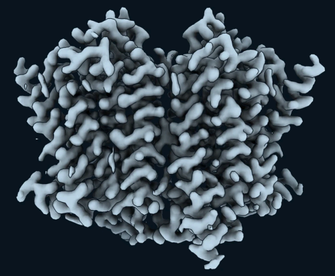
EXPERIMENTAL · EMPIAR-10017 expert coordinates

2D CLASS AVERAGES



RECOMPUTED · simplified classification

3D DENSITY MAP

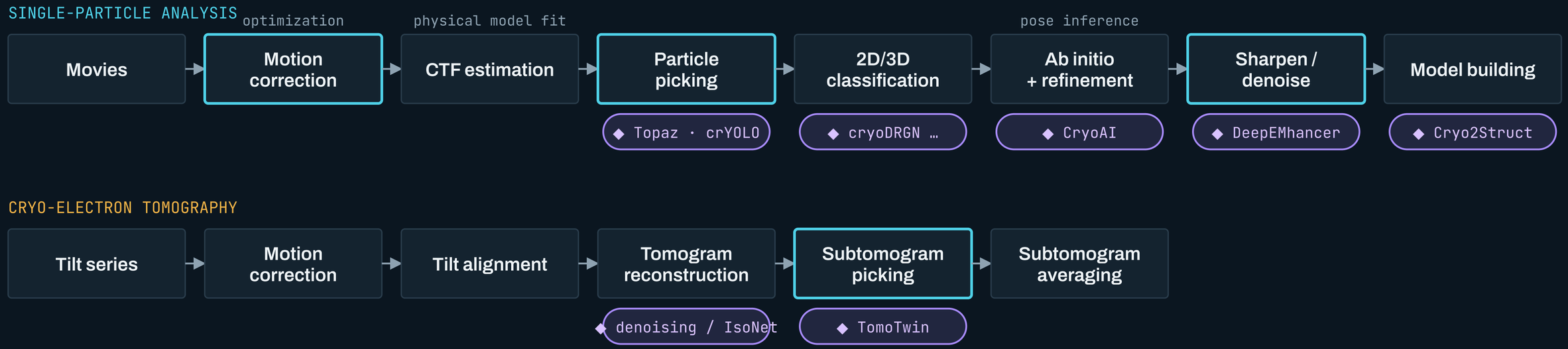


EXPERIMENTAL · EMD-72202

SOURCE Movie: EMPIAR-10204 (β-gal, CRYO ARM 200, 49 frames, 0.885 Å/px). Particles/2D classes: EMPIAR-10017, presenter's simplified classification. Map: EMD-72202 half maps. Tilt series/tomogram: simulation.

▢ CONCEPTUAL SCHEMATIC " REPORTED RESULT

End-to-end workflow — where deep learning has entered



CryoAI (Levy et al. 2022): learned encoder predicts each particle's pose; physics-based decoder; ab initio, homogeneous; reported ≈ one order of magnitude faster on large datasets.

Cryo2StructData (Giri et al. 2024): 7,600 density maps with voxels labelled from atomic structures, to train and test model-building networks.

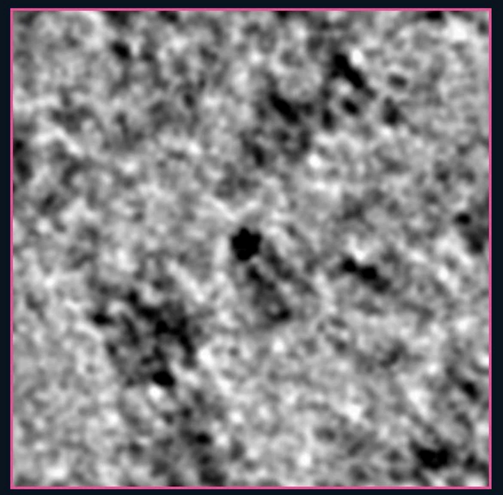
Today's focus (cyan boxes): motion → picking → enhancement — where errors are born and propagate.

SOURCE Iqbal et al. 2026 (workflow, Fig. 1–2, not reproduced) · Levy et al. 2022 ECCV LNCS 13681:540–557 (CryoAI) · Giri et al. 2024 Sci Data 11:458 (Cryo2StructData)

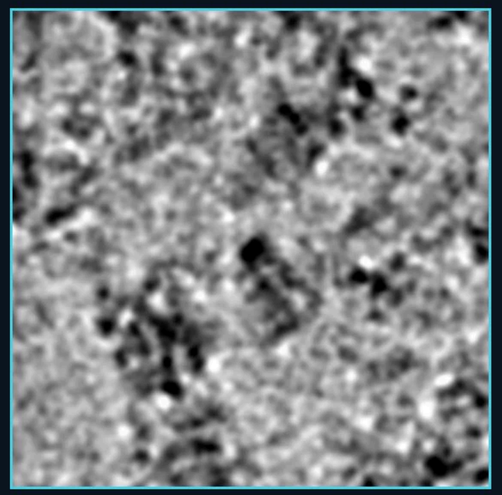
● EXPERIMENTAL DATA ◆ RECOMPUTED RESULT

Motion correction: undoing a 21.3 Å drift

REAL MOVIE, 7 GROUPS OF 7 FRAMES (1.77 Å/PX CROP)



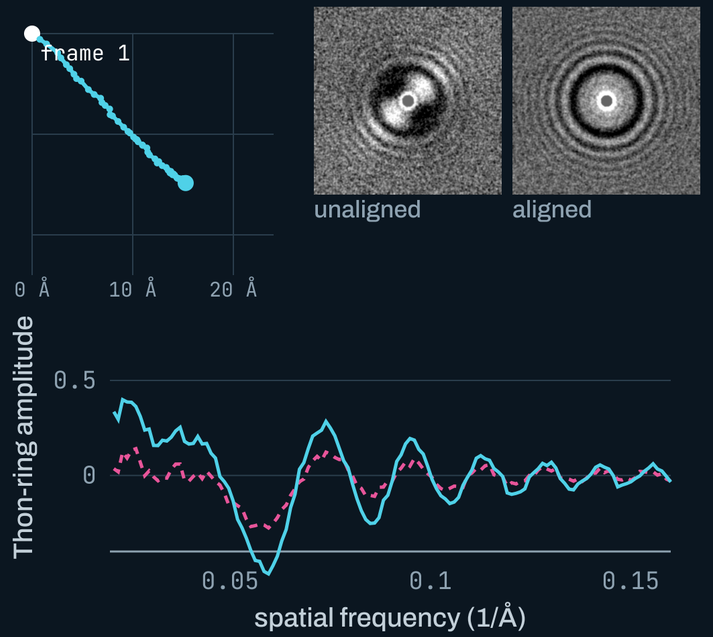
Sum of 49 frames — unaligned



Sum of 49 frames — aligned

Plays once and holds. Drift measured by presenter's cross-correlation alignment (not MotionCor2).

GLOBAL TRAJECTORY · POWER SPECTRA (THON RINGS)



LOCAL PATCHES: RESIDUAL MOTION (×3.2)

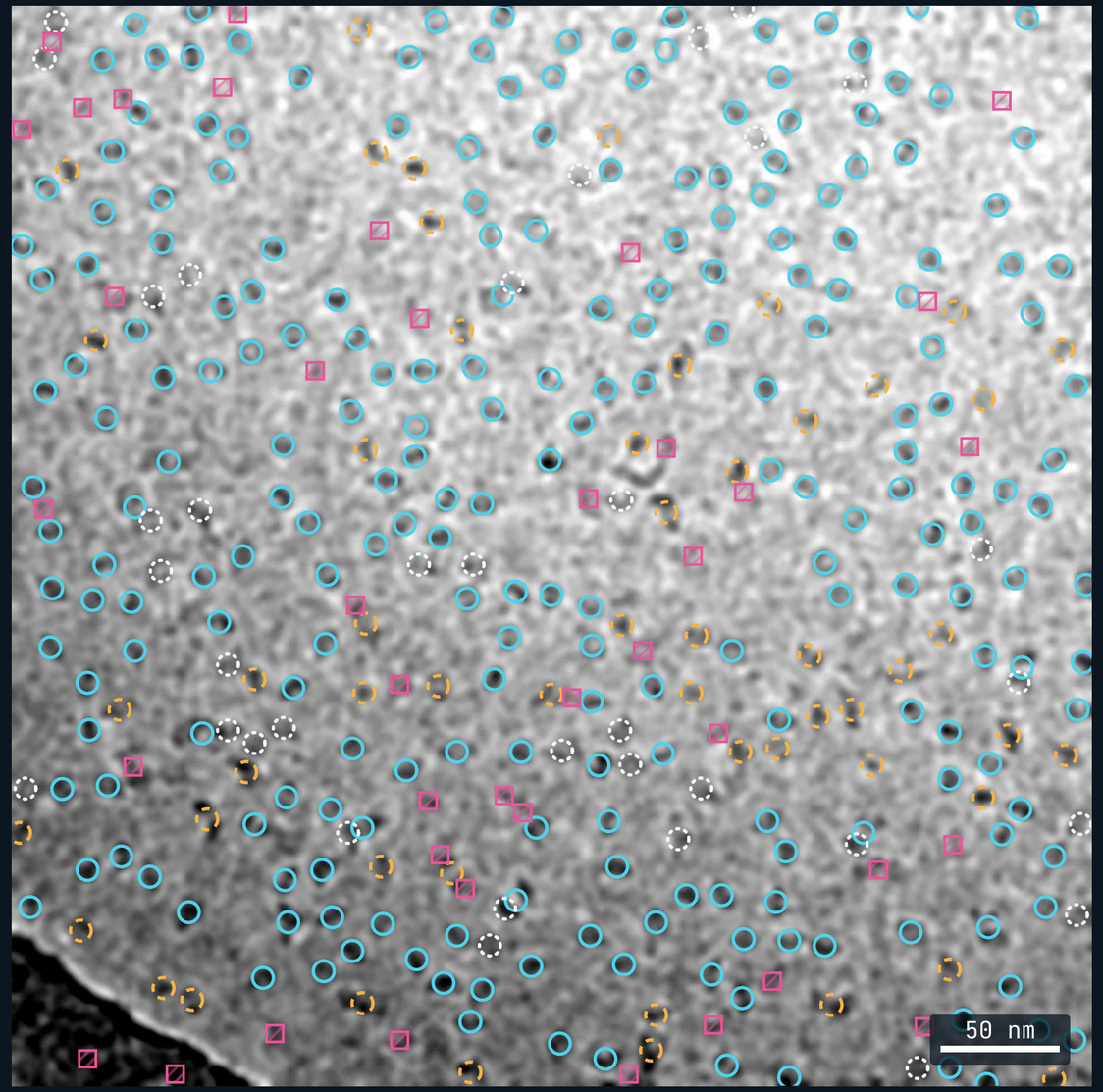


-- corner patch: 10 Å excursion — little signal, probably noise. Bad local estimates propagate into CTF fitting.

MotionCor2, RELION polishing, cryoSPARC patch motion: **conventional optimization, not deep learning.** Dose weighting then down-weights late, damaged frames.

SOURCE EMPIAR-10204 movie 20170629_00021 · presenter's simplified cross-correlation alignment (not MotionCor2) · Zheng et al. 2017 Nat Methods 14:331 · Grant & Grigorieff 2015 eLife 4:e06980

● EXPERIMENTAL DATA ◆ RECOMPUTED RESULT



Particle picking is inference, not detection

- Low contrast: dose-limited SNR
- Ice, carbon edges and contamination look like particles
- Touching, overlapping and aggregated particles
- Variable size, conformational heterogeneity
- Preferred orientation; rare views are scarce
- The "ground truth" is a human judgement

Accepted — model and expert agree (solid) · Uncertain — model only (dashed)

Missed — expert only (dotted) · Rejected — scored below threshold (hatched)

Topaz score threshold = **2.0** (chosen by us, best F1). Over all 12 micrographs: 6,108 agreed · 1,887 model-only · 721 expert-only.

SOURCE EMPIAR-10017 micrograph + expert coordinates · Topaz 0.3.20 pretrained (Bepler et al. 2019 Nat Methods 16:1153), threshold 2.0, 40 px match radius · Dhakal et al. 2025

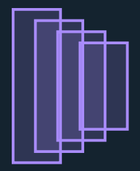
▢ CONCEPTUAL SCHEMATIC " REPORTED RESULT

Topaz, crYOLO and TomoTwin: three learning ideas

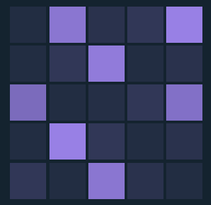
Topaz Bepler 2019



few positives · rest unlabeled
(not negative!)

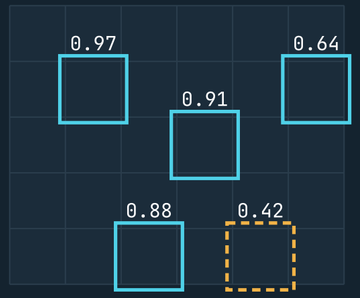


CNN, prior π



score map → threshold
→ 2D coordinates

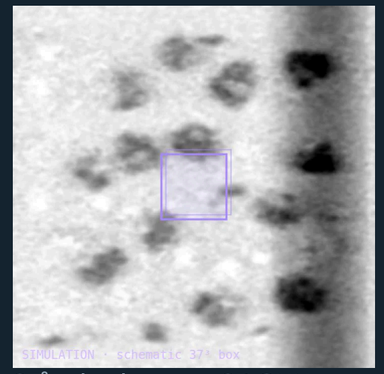
crYOLO Wagner 2019



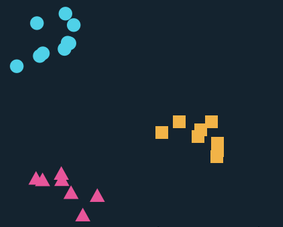
one pass of a CNN:
each grid cell predicts
box + confidence

supervised · YOLO

TomoTwin Rice 2023



37³ subvolumes → 3D CNN



32-D embedding (2-D view)

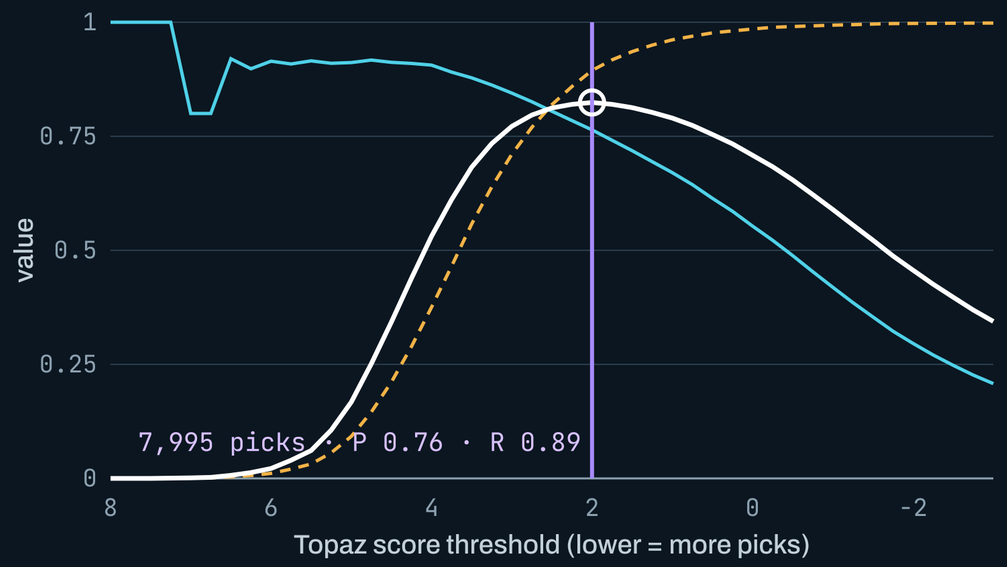
	TOPAZ	CRYOLO	TOMOTWIN
Input	2D micrographs	2D micrographs	3D tomograms (~10 Å/px)
Learning	positive-unlabeled CNN	supervised YOLO detection	deep metric learning (triplets)
Output	score map → 2D coordinates	boxes + confidence	embeddings → 3D coordinates
Labels needed	a few positives per dataset	200–2,500 boxes, or general model	none to retrain: 1 reference or clustering
Best use	large SPA sets, few labels	fast, on-the-fly picking	crowded tomograms, known/related proteins
Main limit	threshold & prior choice; incomplete labels	needs representative training boxes	≳150 kDa; no conformations at picking

SOURCE Bepler et al. 2019 Nat Methods 16:1153 · Wagner et al. 2019 Commun Biol 2:218 · Rice et al. 2023 Nat Methods 20:871

◆ RECOMPUTED RESULT

Reading picker output: metrics first, structure last

TOPAZ (PRETRAINED) VS 6,829 EXPERT PICKS · 12 REAL MICROGRAPHS

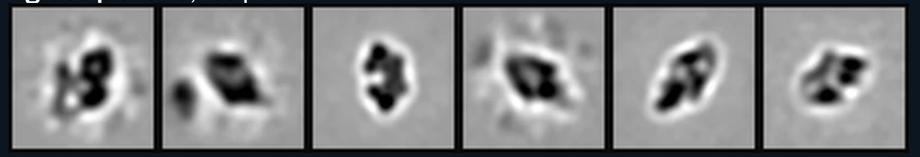


— precision: picks that match expert -- recall: expert picks found — F1 = 2PR/(P+R)

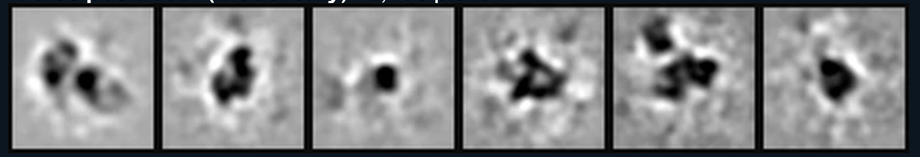
Best F1 **0.82** at threshold 2 (7,995 picks). Topaz retrained on only 25% of labels: F1 **0.86** on held-out micrographs (pretrained 0.82).

SIMPLIFIED 2D CLASSES OF EACH PICK SET

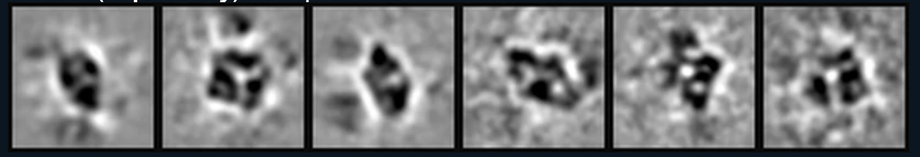
Agreed picks · 6,108 picks



"False positives" (model only) · 1,496 picks



Missed (expert only) · 721 picks



Several "false-positive" classes look like real β-galactosidase views: the labels were incomplete. Largest class = off-centre / overlapping neighbours.

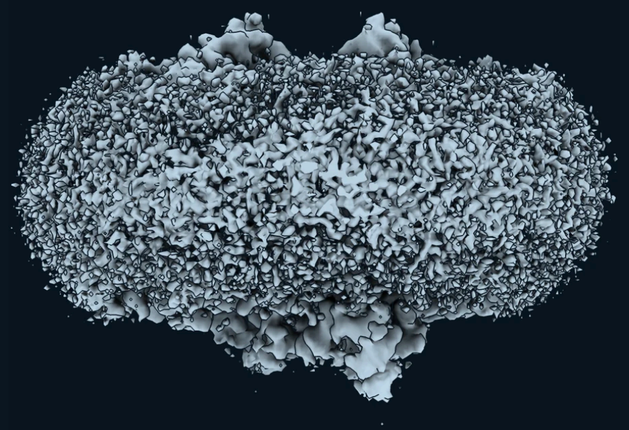
> **Judge downstream:** clean 2D classes · viewing-direction coverage · rare views kept · gold-standard FSC of independent half maps (read at 0.143) · local resolution · is the biology complete? **More particles ≠ better.**

SOURCE Presenter's Topaz 0.3.20 run on 12 EMPIAR-10017 micrographs vs 6,829 expert coordinates (40 px radius) · simplified 2D classification (not RELION/cryoSPARC) · Scheres & Chen 2012 Nat Methods 9:853

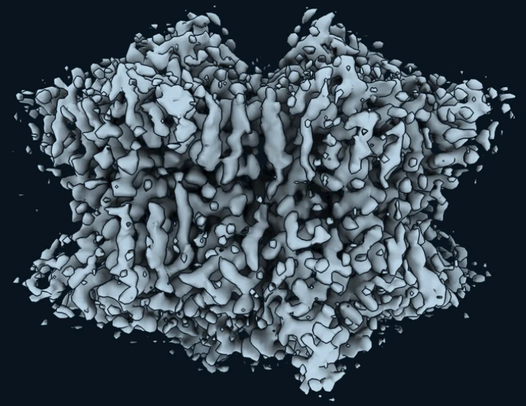
● EXPERIMENTAL DATA ◆ RECOMPUTED RESULT

Map enhancement: cleaner is not the same as more evidence

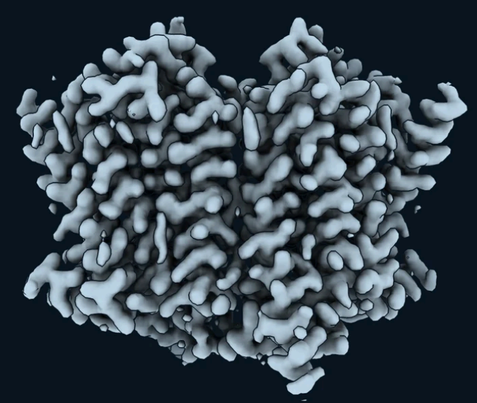
UNSHARPENED AVERAGE OF HALF MAPS



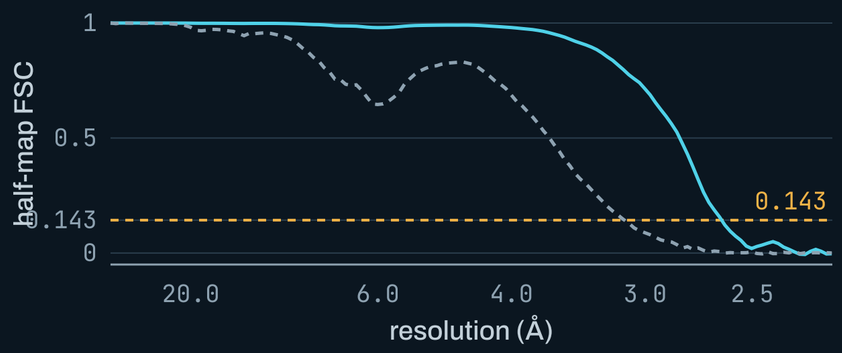
GLOBAL B-FACTOR SHARPENED (B = 66.5 Å²)



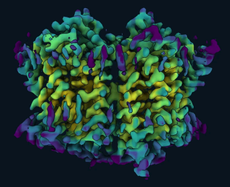
DEEPEMHANCER (TIGHTTARGET)



EMD-72202 (β2AR dimer, nanodisc). Each map contoured where 80% of PDB 9Q3L atoms lie inside **— now at HALF that level: the conventional maps show the lipid nanodisc; DeepEMhancer removed it.**



Independent evidence — unchanged by any post-processing: gold-standard FSC0.143 2.62 Å (soft mask; 3.12 Å unmasked; deposited 2.5 Å). Local resolution 2.39–2.85 Å at the protein atoms, far worse in the nanodisc.



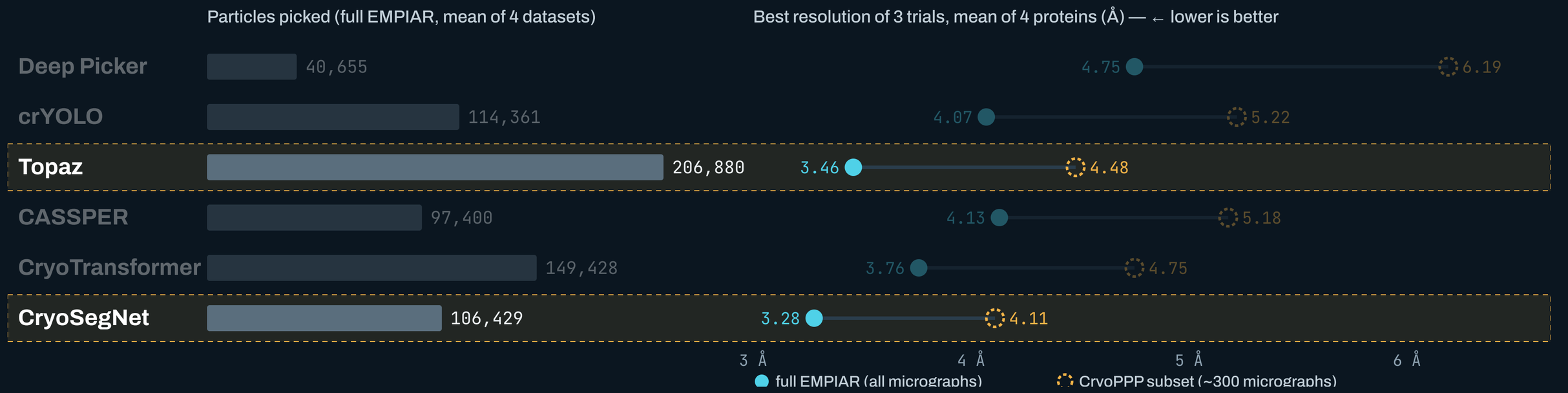


local resolution (Å), viridis; bright = better

SOURCE EMD-72202 half maps (β2AR dimer + AP-7-168) · DeepEMhancer 0.18 (Sanchez-Garcia et al. 2021 Commun Biol 4:874), run by presenter · our FSC, B-factor and local resolution (not cryoSPARC) · Iqbal et al. 2026

“ REPORTED RESULT

Case study: six pickers, same data, same reconstruction protocol



More particles ≠ better map. Topaz picked ~2× CryoSegNet's particles yet averaged 3.46 vs 3.28 Å. Hemagglutinin: 259,757 vs 90,477 particles → 3.21 vs 3.20 Å.

Benchmark-specific: best of 3 seeds; 4 proteins; CryoSegNet not best everywhere — on EMPIAR-10093 Topaz is best in both subsets. Text credits CryoTransformer; Table 3 says Topaz **6.12** vs 6.81 Å (table used). Two winners are from the reviewers' own group.

What explained the differences? The authors point to **viewing-direction coverage**: pickers that missed orientations lost resolution. Validate with orientation plots and half-map FSC, not counts.

SOURCE Dhakal et al. 2025 Brief Bioinform 26:bbaf011, Table 3 — best resolution of 3 cryoSPARC trials, average across 4 test proteins; lower Å is better. Values re-plotted, not reproduced.

▢ CONCEPTUAL SCHEMATIC

When are these tools useful?

● Reach for deep learning when…

Large SPA dataset, few annotations	Topaz (PU learning)
Screening during collection	crYOLO general model, on the fly
Low-contrast, small or irregular particles	learned picker + 2D-class cleanup
Crowded tomograms; known or related complexes	TomoTwin (reference or clustering)
Reproducible high-throughput pipelines	fixed model + logged thresholds
Human picking becomes inconsistent	model as a consistent first pass

◇ Prefer simpler or conventional methods when…

Large, obvious, well-separated particles	blob / template picking is enough
You need a strictly unbiased search	independent methods, compare sets
Data unlike any training set	retrain, or don't trust the scores
Interpretation hinges on weak density	unfiltered half maps, not enhanced maps

Presenter's synthesis of the use cases stated in each method paper — not a published decision tree.

SOURCE Bepler et al. 2019 · Wagner et al. 2019 · Rice et al. 2023 · Dhakal et al. 2025

▢ CONCEPTUAL SCHEMATIC " REPORTED RESULT

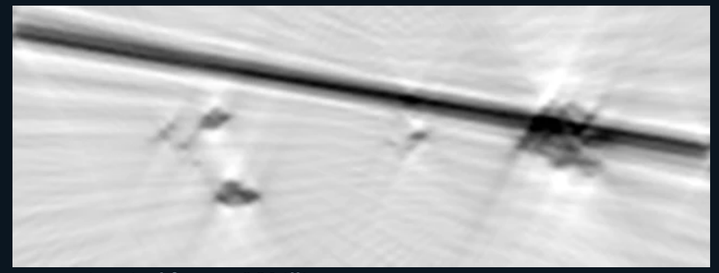
Limitations, artifacts and safeguards

	× FAILURE MODE	✓ SAFEGUARD
DATA	Domain shift: microscope, detector, ice, particle size, preprocessing	Held-out and cross-dataset testing
LABELS	Incomplete / subjective annotations; class imbalance; data leakage	Inspect raw micrographs and 2D classes; split by dataset, not micrograph
PHYSICS	Small particles, overlap, contamination, preferred orientation, missing wedge	Orientation-distribution and local-resolution analysis
MODEL	Overconfident scores; threshold sensitivity; over-sharpening; invented or erased density	Independent half maps the network never saw; compare ≥2 methods
EVALUATION	Authors benchmark their own tools; best-of-N reporting; compute cost	Biochemical validation (mutants, ligands, assays); report uncertainty

MISSING WEDGE · X–Z SLICE · SIMULATION



truth (simulated volume)



reconstructed from ±60° tilts



full 180° (impossible in practice)

±60° data smear densities along z and add X-shaped streaks: similarity searches see shapes that depend on orientation.

SOURCE Iqbal et al. 2026 · Dhakal et al. 2025 · Rice et al. 2023 · Sanchez-Garcia et al. 2021 · Scheres & Chen 2012

Take-home

1. **AI is most valuable when it removes repetitive inference** — picking, triage — while preserving the experimental evidence.

2. **A visually cleaner result is not automatically a more accurate one** — more picks, prettier maps and higher confidence can all mislead.

3. **Validate every AI output at the level of the biological question** — half maps, viewing directions, local resolution at the site you care about.

The goal is not to maximize picks or make the prettiest map.
The goal is to preserve structural information — and know when the model is uncertain.

Key references · Questions?

1. Iqbal et al. (2026). Artificial intelligence in cryo-EM: emerging deep neural network methods from sample preparation, particle picking, map reconstruction, modelling to enhanced resolution. BMC Artificial Intelligence 2:2. doi:10.1186/s44398-025-00017-2
2. Dhakal et al. (2025). Artificial intelligence in cryo-EM protein particle picking: recent advances and remaining challenges. Briefings in Bioinformatics 26:bbaf011. doi:10.1093/bib/bbaf011
3. Levy et al. (2022). CryoAI: Amortized Inference of Poses for Ab Initio Reconstruction of 3D Molecular Volumes from Real Cryo-EM Images. Computer Vision – ECCV 2022:540-557. doi:10.1007/978-3-031-19803-8_32
4. Giri et al. (2024). Cryo2StructData: A Large Labeled Cryo-EM Density Map Dataset for AI-based Modeling of Protein Structures. Scientific Data 11:458. doi:10.1038/s41597-024-03299-9
5. Bepler et al. (2019). Positive-unlabeled convolutional neural networks for particle picking in cryo-electron micrographs. Nature Methods 16:1153-1160. doi:10.1038/s41592-019-0575-8
6. Wagner et al. (2019). SPHIRE-crYOLO is a fast and accurate fully automated particle picker for cryo-EM. Communications Biology 2:218. doi:10.1038/s42003-019-0437-z
7. Rice et al. (2023). TomoTwin: generalized 3D localization of macromolecules in cryo-electron tomograms with structural data mining. Nature Methods 20:871-880. doi:10.1038/s41592-023-01878-z
8. Sanchez-Garcia et al. (2021). DeepEMhancer: a deep learning solution for cryo-EM volume post-processing. Communications Biology 4:874. doi:10.1038/s42003-021-02399-1
9. Zheng et al. (2017). MotionCor2: anisotropic correction of beam-induced motion for improved cryo-electron microscopy. Nature Methods 14:331-332. doi:10.1038/nmeth.4193
10. Shen et al. (2026). A biased allosteric modulator is a molecular glue for β2AR dimerization. Nature 658:556-564. doi:10.1038/s41586-026-10892-y
11. Scheres (2015). Semi-automated selection of cryo-EM particles in RELION-1.3. Journal of Structural Biology 189:114-122. doi:10.1016/j.jsb.2014.11.010
12. Scheres & Chen (2012). Prevention of overfitting in cryo-EM structure determination. Nature Methods 9:853-854. doi:10.1038/nmeth.2115

Data: EMPIAR-10017, EMPIAR-10204, EMD-72202 (public). Tools run by the presenter: Topaz 0.3.20, DeepEMhancer 0.18, ChimeraX 1.12. Claim ledger, BibTeX and asset provenance ship with the deck.

Full bibliography (25)

SOURCE Full bibliography: references.bib · claims: src/data/claims.yaml · provenance: provenance/assets.json